

Behavioral responses in the top-tax project

A methodology

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Tax-Simulator

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Abstract

This paper documents the behavioral responses and cross-base interactions used in the top-tax project, which estimates the additional revenue available from the top of the income and wealth distribution. It describes each channel under a common template—mechanism, parameters, calibration, interactions, and limitations—and the architecture that lets the channels interact on shared record-level state. This is a working draft: the capital-gains realization channel (Section 4.1.1 and Appendix A.1) is complete and serves as the template exemplar; the remaining channels are drafted or outlined to the same structure.

Contents

1	Introduction	1
2	Model structure	2
2.1	Static and conventional passes	2
2.2	Two channel types	2
2.3	Run order	2
3	Interaction architecture	2
3.1	Shared objects	2
3.2	The reporting–real separation	3
3.3	Composition of the taxable-income elasticity	3
4	Behavioral channels	3
4.1	Realization and timing at death	4
4.2	Taxable-income base-shifting	6
4.3	Wealth and estate own-rate responses	7
4.4	Saving and financing	7
4.5	Corporate incidence and labor-market margins	8
5	Limitations and open items	8
6	References	9

A	Derivations and calibration	9
A.1	Capital-gains realization	9
B	Parameter and provenance table	10

1 Introduction

Several current proposals draw revenue from the same taxpayers through distinct statutes: top ordinary rates, capital-gains rates, the treatment of gains at death, the estate tax, a wealth tax, the corporate tax, and repeal of the qualified business income deduction. Scoring each in isolation and summing overstates their combined yield, for two reasons.

First, the bases overlap mechanically. A given stock of wealth is capital income during life, an estate at death, a wealth-tax base in the interim, and corporate profit upstream. A dollar collected under one statute is unavailable to another, so summing standalone estimates double-counts it.

Second, taxpayers respond. They retime realizations, convert ordinary income into deferred gains, shift income across the corporate and pass-through boundary, conceal assets, and finance tax payments out of saving. These responses move income between bases and out of the taxable total, and a response induced by one instrument alters the base on which the others operate.

The model estimates these effects on record-level balance sheets rather than through an assumed aggregate elasticity. Each channel described below is a structural response measured against the microdata; the object of interest is their interaction (Section 3), which is why the project reports the conventional estimate as a share of the static estimate and a by-base destination ledger rather than a single headline elasticity.

Two of the channels documented here, the wealth-financing channel (Section 4.4.1) and corporate incidence (Section 4.5.1), are mechanical rather than elasticity-driven. They are included because they transmit a policy’s effect from one base to another, which is the interaction the project measures.

2 Model structure

2.1 Static and conventional passes

Each non-baseline scenario is run twice. The *static* pass holds taxpayer inputs at baseline levels and recomputes tax under the reform; it is the law-only counterfactual. The *conventional* pass activates the behavioral channels. The reported quantities are differences between the two: the static estimate is the mechanical intent, and the conventional estimate is the amount that survives behavior. Interactions are reported through receipts as static minus conventional deltas. Distribution tables are static-sourced by convention.

2.2 Two channel types

The model implements behavior in two forms.

Type	Implementation
Behavior module	An R file at <code>config/scenarios/behavior/{var}/{name}.R</code> exposing a <code>do_{var}()</code> hook, dispatched on the behavioral pass. It modifies tax-unit inputs; tax is then recomputed. Examples: realization, income conversion, entity shifting, evasion, wealth avoidance, charity, employment.
Mechanical applier	A fixed step invoked at the head of the conventional pass in <code>src/sim/run.R</code> , with no hook and no elasticity dispatch. Examples: the wealth-financing channel and corporate incidence.

2.3 Run order

Because the channels touch overlapping bases, their order is part of the specification. The conventional sequence is

corporate incidence $\rightarrow$ wealth haircut $\rightarrow$ $\underbrace{\text{realization} \rightarrow \sigma \rightarrow \text{entity shifting} \rightarrow \text{evasion}}_{\text{fixed module order}} \rightarrow \text{do_taxes}.$

The module order is asserted at runtime so that each dollar is adjusted at most once: realizations are set before conversion prices them, conversion before the entity margin re-routes the remainder, and the reporting channels (evasion, avoidance) run last, after which only `do_taxes` reads the frame (Section 3.2).

3 Interaction architecture

Three structures let the channels interact on common state rather than through hand-set cross-elasticities.

3.1 Shared objects

Several channels are priced from the same internal quantities:

- $\tau_{eq}(a, t)$, the present value of tax per dollar entering the unrealized-gain state, produced by the realization machinery (Section 4.1.1). It prices the conversion exit (Section 4.2.1) and, on the roadmap, the entity-shifting deferral value (Section 4.2.2).
- F , the death-forgiveness value of an unrealized gain, which depends on the realization-at-death regime and on estate exposure (Section 4.1.3). It links realization to the estate tax.
- $h = \tau_w \tau_{cg}$, the wealth-tax carry on a deferred gain (Section 4.1.2), which couples a wealth tax into the realize-or-hold decision.

Because these objects are shared, a change to one instrument propagates to the others through the model's own state.

3.2 The reporting–real separation

The governing rule is that cash is reported and balance sheets are real. A reporting response (evasion, wealth avoidance, estate concealment) affects only the cash tax paid, and through it the financing flow of the wealth channel (Section 4.4.1). It does not affect the economic balance sheet (`value.*`), the gain stock, cohort assignment, heir inheritances, or corporate exposure. A

real response (corporate markdown, wealth haircut, realization, conversion, entity shifting) flows into every base, including reported ones.

The architecture enforces this. Every real-side consumer reads raw input data, static detail, or a preserved raw column, never the behavior-modified conventional frame. Reporting modules write only to tax-computation inputs and to documented isolation points (the materialized `net_worth` column, the `estate_concealed_frac` input to `calc_estate`). This is why the reporting modules are ordered last (Section 2.3). Each channel below is marked REAL or REPORTING.

3.3 Composition of the taxable-income elasticity

The top-bracket elasticity of taxable income (ETI) is composed from three modules—income conversion (Section 4.2.1), entity shifting (Section 4.2.2), and evasion (Section 4.2.3)—together with charity. One parameter, σ , is calibrated as a residual so that the modeled top-ordinary ETI matches an external target of approximately 0.25, conditional on the rest of the stack.

A consequence is that any change to a margin inside the ordinary-rate ETI bundle requires σ to be re-derived. Margins outside the bundle—the wealth and estate terms in the realization problem, the estate own-rate response, and capital-gains realization itself—do not affect σ . This distinction governs calibration maintenance and is monitored by an automated staleness check (Section 5).

4 Behavioral channels

Each channel is documented under seven headings:

1. *What it captures* — the economic response, stated briefly.
2. *Mechanism* — how the model produces it.
3. *Type and location* — module or mechanical; REAL or REPORTING; code path.
4. *Parameters* — the pinned values and override switches.
5. *Calibration and evidence* — how the parameters are set and the literature anchor.
6. *Interactions* — the other channels and bases it touches.
7. *Limitations* — what it omits.

Medium-depth content appears here; full derivations are in Appendix A.

4.1 Realization and timing at death

4.1.1 Capital-gains realization: the entropy Bellman

What it captures. The share of a mechanical capital-gains rate increase that survives once realizations respond. Investors defer gains when rates rise, and the value of deferral depends on the treatment of the gain at death. The channel models both the long-run level of realizations and the short-run retiming around a rate change.

Mechanism. Investors are grouped into representative cells defined by age cohort and within-age net-worth percentile. For each cell the model solves a realize-or-hold problem over the gain stock under an entropy realization cost. Each dollar of unrealized gain carries a non-tax motive to sell—liquidity, rebalancing, or consumption—drawn from an exponential reservation

distribution; the holder realizes the dollar when that motive exceeds the tax wedge between realizing now and deferring. The realization rate is the survival function of that distribution. The first-order condition is

$$r_D^j = r_D^B \exp(-\eta(MC^j - MC^B)), \quad \text{clipped to } [0, 1], \quad (1)$$

where the marginal cost of holding rather than realizing is

$$MC = \tau + \beta(1 - m) W_{\text{next}} + \beta m F, \quad F = (1 - c_\phi) \tau. \quad (2)$$

Here τ is the current-year rate, W_{next} the discounted continuation value of an unrealized dollar under survival probability $1 - m$, and F the discounted death-forgiveness value under mortality m . The realization-at-death regime enters through c_ϕ , the share of the gain's tax burden the holder internalizes at death: $c_\phi = 0$ under step-up ($F = \tau$), $c_\phi = 1$ under deemed realization ($F = 0$), and an intermediate value under carryover.

Since $d \ln r_D / dMC = -\eta$ across the whole pool, η is the long-run realization semi-elasticity directly, with no inert floor or responsive-share deflation; this is the single-pool property of the current specification (Appendix A.1). A separate short-run overlay lets a calibrated fraction of realizations retime by one year toward the lower-wedge year. The overlay nets to zero under a uniform permanent shock and therefore leaves the long-run response unchanged.

Type and location. Behavior module. The applicer `do_kg_dynamics` (`config/scenarios/behavior/kg_dynamics/turnover.R`) is an allocator that reads per-year cell state and translates it to per-record `kg_lt` adjustments. State is produced by a pre-pass (`kg_dyn_run_bathtub_pass`) in `src/sim/kg_dynamics.R`; shared cohort primitives are in `src/sim/cohort_bathtub.R`. REAL.

Parameter	Value	Meaning	Override
η	2.4825	long-run realization semi-elasticity	KG_ETA
timeable share	0.2542	fraction of realizations that retimes by one year	KG_TIMEABLE_SHARE
deemed-avoidance	0.25	valuation discount on deemed realizations (data, not law)	KG_DEEMED_AVOIDANCE
c_ϕ	0/ θ /1	regime death-burden share	—
spec version	3	single pool	—

Parameters.

Calibration and evidence. η is pinned on the full simulator. The calibration harness measures the realized elasticity $E_{\text{full}}(\eta) = d \log R / d\tau_{rw}$ at simulation year 30 across a grid of η values; the surface is linear through the origin, so a few points determine it. Inverting the fitted line for the target $E_{\text{full}}^* = -0.6/0.238 = -2.52$ (a top-rate realization elasticity over the top-rate divisor, following the JCT and CBO convention and the estimates of Dowd, McClelland, and Muthitacharoen) at slope 1.0155 gives $\eta^* = 2.4825$. The fitted coefficients are recorded so that the next re-pin is arithmetic. The timeable share is calibrated separately against the short-run announcement moment given η . The full procedure is in Appendix A.1.

Interactions. The channel produces τ_{eq} and F (Section 3.1), which price income conversion (Section 4.2.1) and entity shifting (Section 4.2.2). The death regime links realization to the estate tax: closing step-up raises realizations. The wealth-carry term (Section 4.1.2) and the estate death-value offset (Section 4.1.3) enter MC and F , coupling the wealth and estate instruments into the realization decision.

Limitations. The five tracked wealth classes are collapsed into one asset bucket; per-asset-class disaggregation is on the roadmap. The revenue-maximizing grid argmaxes at the current boundary (a statutory peak near 44–45%) and should be extended to identify the turning point. The deemed-avoidance discount is a data calibration consistent with JCT scoring, not a behavioral response.

4.1.2 Wealth-carry

What it captures. A wealth tax raises the cost of deferring a gain, because the unpaid capital-gains tax on an unrealized gain remains in the wealth-tax base and is taxed at τ_w each year it is deferred. This unlocks realizations in wealth-taxed cells that would otherwise defer, and it lowers the value of the conversion deferral path (Section 4.2.1).

Mechanism. A per-year carrying cost $h = \tau_w \tau_{cg}$ is added to the marginal cost of holding in (2) and to the τ_{eq} recursion. It is computed at the record level and then aggregated, not as a product of separately averaged rates, because τ_w and τ_{cg} are positively correlated at the top. The term applies on the survival branch only; in-year realizers and decedents pay no carry that year.

Type and location. A structural term inside the realization module (`src/sim/kg_dynamics.R`). REAL.

Parameters. None free. h is the product of two marginal rates already in the model, τ_w from `mtr_net_worth` and τ_{cg} from `mtr_kg_lt`. Under current law $\tau_w = 0$ for essentially all cells, so $h = 0$ is an exact no-op, which is why the term did not change the specification version or the η calibration.

Calibration and evidence. No separate calibration. The magnitude follows from the deferral arithmetic: the per-year benefit of deferral is approximately $r_{\text{real}} \tau$, so a one-percent wealth tax offsets roughly half of it at $r_{\text{real}} \approx 2\%$.

Interactions. Couples the wealth-tax instrument into realization and into the conversion price.

Limitations. The response is aggregated at the age-cell level rather than the record level, which understates the record-level response by Jensen’s inequality (conservative, never an overstatement); the measured understatement ranges from about 8% to 33% depending on the parameter corner. An exposed/unexposed split closing the gap was considered and not adopted. The understatement should be noted when reporting wealth and capital-gains corners jointly.

4.1.3 Estate death-value offset

What it captures. For an estate-taxable decedent, capital-gains or deemed tax paid at or before death reduces the taxable estate, so the effective rate near death is approximately $\tau_{cg}(1 - \tau_e)$. Including this stops the model from overstating lock-in for estate-exposed cells when the deemed and estate switches are active. *[Draft pending.]*

Mechanism, parameters, and limitations. Gain-weighted cell aggregation of a switch-gated marginal estate rate entering F and MC ; baseline death value $F_B = \tau_B(1 - e_B)$, paired by leg. Inside the realization module; REAL. *[Draft pending.]*

4.2 Taxable-income base-shifting

4.2.1 Income conversion (σ)

What it captures. Conversion of labor and ordinary income into the deferred unrealized-gain state (the founder-equity and carried-interest path), a component of the top-ordinary ETI. *[Draft pending.]*

Summary. Module `do_conversion` (`config/scenarios/behavior/conversion/sigma.R`; shared function `src/sim/sigma_conversion.R`); REAL. Parameter $\sigma = 0.16$ (SIGMA_CONV), re-derived in 2026, superseding an earlier 0.08. Calibrated as a residual to a top-ordinary ETI of 0.25 (Section 3.3). Consumes τ_{eq} ; a change to the entity, evasion, or charity margins requires re-derivation. The conversion path is currently priced free of the corporate layer. *[Draft pending.]*

4.2.2 Entity shifting

What it captures. Business income relocating across the corporate and pass-through boundary in response to the corporate-to-individual rate differential, following Pearce and Prisinzano. *[Draft pending.]*

Summary. Module `do_entity_shifting` (`config/scenarios/behavior/entity_shifting/pearce_prisinzano.R`); REAL. The shifting semi-elasticity is presently a fixed value of 0.25. This is a known limitation: it should be derived from the model's own deferral value τ_{eq}/τ , which would make the corporate retention shelter respond to the realization-at-death regime rather than remaining invariant to it. *[Draft pending.]*

4.2.3 Evasion

What it captures. Under-reporting of self-employment, partnership, S-corporation, and rental income that rises with the net-of-tax rate; the leakage component of the ETI. *[Draft pending.]*

Summary. Module `do_evasion` (`config/scenarios/behavior/evasion/debacker.R`); REPORTING. Net-of-tax elasticities: Schedule C/F 0.046, partnership and S-corporation 0.052, rent 0.040. Wages, interest, and dividends do not respond, reflecting information reporting. Evidence from DeBacker, Heim, and Yuskavage. Feeds the hidden-ledger link (Section 4.3.1). Limitations: positive legs only; the overstated-loss margin and a top-graded multiplier are deferred. *[Draft pending.]*

4.3 Wealth and estate own-rate responses

4.3.1 Wealth avoidance and the hidden ledger

What it captures. Reduction in reported taxable wealth under a wealth tax, through legal valuation management and concealment, with concealment made consistent across bases so that

a concealed dollar also escapes income, capital-gains, and estate tax. *[Draft pending.]*

Summary. Module `do_wealth` (`config/scenarios/behavior/wealth/avoidance.R`); `REPORTING`, writing only the `net_worth` isolation point and `estate_concealed_frac`. Semi-elasticities: marketable wealth -7 , closely-held wealth -17 . Concealment shares: public 1.0, private 0.5. Migration is subsumed here as a ceiling (Section 4.5.2). *[Draft pending.]*

4.3.2 Estate own-rate response

What it captures. Reduction in reported gross estate in response to a change in the net-of-estate-tax rate. *[Draft pending.]*

Summary. Part of `do_wealth`; `REPORTING`, via `estate_concealed_frac`. Elasticity $\varepsilon = 0.16$ (`ESTATE_REPORT_EPS`; Kopczuk–Slemrod band 0.10–0.22), in the net-of-tax form retained $= ((1 - \tau_S)/(1 - \tau_B))^\varepsilon$, keyed to the change in the marginal estate rate rather than its level. Evidence from Kopczuk and Slemrod. A rate-responsive charitable-bequest margin is not yet built. *[Draft pending.]*

4.4 Saving and financing

4.4.1 The wealth-financing channel

What it captures. A share $s = 1 - \text{MPC}$ of the net above-baseline during-life tax is financed out of wealth rather than consumption. It compounds over time and enters the estate and capital-income bases at death. This is the channel through which a during-life tax reduces future wealth-based revenue. *[Draft pending.]*

Mechanism. A cohort recurrence with kernel and forcing

$$G(a, p, t) = (1 + r_{\text{total}}(t)) - s(\tau_y + \tau_w), \quad F = \Delta T^0 - \Delta Y_{\text{exog}}.$$

[Draft pending.]

Summary. Mechanical applier (`src/sim/wealth_dynamics.R`, `src/sim/cohort_bathtub.R`), run at the head of the conventional pass; `REAL`. Parameters: the saving-share profile $s(\text{age}, \text{net-worth percentile})$, calibrated from about 0.1 at the bottom to 0.80 at the top; a transition matrix; a cap `fmax` = 0.9; and r_{total} equal to nominal GDP-per-capita growth. It is a financing response, not a rate-of-return saving elasticity. It requires a dedicated pass, roughly doubling compute. *[Draft pending.]*

4.5 Corporate incidence and labor-market margins

4.5.1 Corporate incidence

What it captures. The record-level incidence of a corporate rate change: cuts to dividend, interest, rent, and pass-through flows; an equity markdown on exposed asset stocks; a capital-gains adjustment; dissaving through the wealth channel; and an endogenous individual-tax offset. *[Draft pending.]*

Summary. Mechanical, revenue-side applier (`src/sim/corp_incidence.R`) with fail-closed activation on `corporate_meta.yaml`; REAL. Parameters are hardcoded `CORP_*` constants, several of which are placeholders pending measurement and disclosed as such. The endogenous offset makes the corporate row sensitive to stacking order. The most complete existing documentation is `other/corporate_incidence/CONSIDERATIONS.md` and `FORMAL_MODEL.md`. *[Draft pending.]*

4.5.2 Charity, employment, and migration

- **Charity.** Module `do_charity`; tax-price elasticity -0.5 on cash giving. The appreciated-asset margin, through which a higher capital-gains rate raises the subsidy to donating securities, is not built. *[Draft pending.]*
- **Employment.** Module `do_employment` (`config/scenarios/behavior/employment/bastian.R`); extensive-margin wage elasticities following Bastian. Second-order for the top-tax question; documented for completeness. *[Draft pending.]*
- **Migration.** No standalone channel. It is subsumed as a ceiling within the wealth-avoidance semi-elasticity and disclosed as such.
- **Excluded by convention.** Real labor supply and a rate-of-return saving elasticity are excluded; the wealth channel is financing, not saving.

5 Limitations and open items

Two facts frame this section. First, the realization problem prices from a single tax rate plus the shared F , h , and estate terms, so any mispriced cross-rate content appears as a missing interaction rather than as noise. Second, σ is a residual, so a gap inside the ETI bundle carries a re-calibration cost that a gap outside it does not (Section 3.3).

Item	Ref.	Effect	Status
Entity-shifting fixed at 0.25	4.2.2	shelter invariant to death regime	open, in-bundle
Corporate constants placeholder	4.5.1	corporate estimate and ETR band	disclosed
Wealth-carry cell aggregation	4.1.2	understates by 8–33%, conservative	accepted
Appreciated-asset giving margin	4.5.2	understates giving and lock-in at high rates	open, in-bundle
Migration not explicit	4.5.2	subsumed as a ceiling	by convention
Corporate layer absent from σ	4.2.1	conversion priced corporate-free	open, in-bundle
Estate own-rate coverage	4.3.2	no charitable-bequest response	partial
Evasion coverage	4.2.3	positive legs only; flat top multiplier	partial

Calibration maintenance. The residual parameters, σ and η under its conditioning assumptions, are monitored by an automated check: a low-cost internal-moment drift test on each run, and a σ re-derivation triggered by any commit that touches the conditioning set (entity shifting, evasion, the charity elasticity, the conversion pool, or the input-data vintage). Reference values are stored in `other/kg_model_tests/calibration_reference.csv`. The check detects drift; it does not re-pin automatically.

6 References

To be completed. Kopczuk and Slemrod (2001); Dowd, McClelland, and Muthitacharoen (2015); DeBacker, Heim, and Yuskavage; Pearce and Prisinzano (2018); Bastian (2023); Saez and Zucman on wealth-tax avoidance; and the JCT and CBO realization-elasticity conventions.

A Derivations and calibration

A.1 Capital-gains realization

The problem. For a representative cell, let r_D be the discretionary realization rate on the gain pool. Each unrealized dollar carries a non-tax reservation benefit $b \geq 0$ to selling, drawn from an exponential distribution, and the holder realizes when b exceeds the tax wedge between realizing now and holding. The entropy realization cost $C(r_D)$ is the cost whose marginal is the inverse survival function of that exponential,

$$C'(r_D) = \frac{1}{\eta} \ln(r_D/r_D^B), \quad C'(r_D^B) = 0, \quad (3)$$

so the baseline rate is the zero-cost reference. The holder equates the net benefit of realizing, $\kappa - MC$, to $C'(r_D)$, with κ the reservation constant and MC as in (2).

Two passes. Baseline inversion gives $C'(r_D^B) = 0$, hence $\kappa = MC^B$ exactly, interior and at the corner. The scenario condition $\kappa - MC^j = C'(r_D)$ then yields (1). Only the upper clip binds, and cells with $r_D^B = 0$ stay at zero. Since $d \ln r_D / dMC = -\eta$ everywhere and the whole pool responds, η is the aggregate long-run semi-elasticity and the revenue-maximizing rate is approximately $1/\eta$.

Single-pool specification. An earlier specification split baseline realizations into a responsive share and an inert share, the latter a point mass of fully constrained sellers. That floor was an artifact of an earlier quadratic cost with no tail, and it removed the interior revenue maximum. The exponential reservation distribution already represents the heterogeneity the floor stood in for, so the current specification drops it: one pool, with a level margin from the Bellman and a timing margin from the overlay.

Timing overlay. A fraction f of baseline realizations retimes across a one-year window toward the lower-wedge year, with move-share clamp($\Delta\text{wedge}/\text{ref_wedge}, 0, 1$) and $\text{ref_wedge} = 0.05$. Composed as a net shift $r_S = r_{\text{ord},S} + (r_{\text{plan},S} - r_{\text{plan},B})$, it nets to zero under a uniform permanent shock, so η is identified independently of f .

Calibration of η . On the full simulator, measure $E_{\text{full}}(\eta) = \log(R_{\text{shock}}/R_{\text{base}})/\Delta\tau_{rw}$ at simulation year 30 across a grid; the surface is linear through the origin. Invert for $\eta^* = |E_{\text{full}}^*|/\text{slope}$ with $E_{\text{full}}^* = -2.52$ and slope 1.0155, giving $\eta^* = 2.4825$ (grid $\eta \in \{2, 2.3992, 3\}$ mapping to $E_{\text{full}} \in \{-2.05, -2.44, -3.03\}$). Provenance is guarded: a stale value warns, and a strict-calibration flag halts.

Appendix derivations for the remaining channels follow at this depth.

Constant	Value	Reference moment	Derived
KG_DYN_DEFAULT_ETA	2.4825	full-sim $E_{\text{full}} = -2.52$ / slope 1.0155	2026-07-12
KG_DYN_TIMEABLE_SHARE	0.2542	full-sim short-run moment	2026-07-09
SIGMA_CONV	0.16	top-ordinary ETI ≈ 0.25	2026-07-12
ESTATE_REPORT_EPS	0.16	Kopczuk–Slemrod (band 0.10–0.22)	2026-07-12
Evasion (C/PT/rent)	0.046/0.052/0.040	DeBacker–Heim–Yuskavage	—
Wealth avoidance (mkt/closely-held)	−7/ − 17	author-accepted	2026-07-08
Concealment share (public/private)	1.0/0.5	consistency across bases	2026-07-08
Entity shifting	0.25 (stub)	should be τ_{eq}/τ	—

B Parameter and provenance table

The reference-moment column is also the source for the staleness check (`calibration_reference.csv`).